



HABIB UNIVERSITY

DHANANI SCHOOL OF SCIENCE AND ENGINEERING

EE251/CE252	Spring'26	Course Project	Weight 15% Max. Marks:100	Due Date: As Announced
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CLO-5	Investigate complex engineering problems in signals and systems and formally present the results.	PLO-4 (Investigation)	Cognitive Level-5 (Evaluating)
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Instructions:

This is a group-based project; you are required to make a group of three students. The project is divided into two milestones:

Milestone 1 [50 Marks] – Submission due during WK#12

Complete Task A mentioned below. To assess this milestone, there will be a group-level viva during Semester Wk#12. Students are required to lock a 30 min viva with the instructor.

The viva will focus on the project code and the associated results. The instructor may ask questions to any member of the group; therefore, all group members are expected to understand every aspect of the project milestone deliverables.

Here is the grading rubric:

1. Milestone completion: 20 (group level)
2. Individual contribution/viva: 20
3. Well-written code and short report submission on results/analysis: 10 (group level)
The report should provide a concise discussion analyzing the results and insights from Parts 1 through 7 of Task A.

Milestone 2 [50 Marks] – Code and report submission due WK#15

Complete task B. Here is the grading rubric:

1. Selection of the signal data sample: 05 (individual level)
2. Signal data analysis and code: 15 (individual level)
3. Well-written code and short report submission on results/analysis: 10 (group level)
The report should provide a concise discussion analyzing the results and insights from Parts a through d of Task B.
4. Class Presentation: 20 (group level) – WK#16

Course Project Problem - *Face the Music*

The purpose this project is to familiarize you with the process of studying and solving real-life engineering problems related to Signals & Systems from real measurements and simulated effects.

1. Preparations

- Carefully read through the entire problem and discuss your strategy with your team-mates.
- Document your solutions, codes, work distribution, and ways-of-working properly, as you will need them for the presentation.

2. Student in a symphony orchestra – Keynotes and Overtones

The sound from most acoustic instruments consists of a fundamental frequency, often termed a keynote, and some overtones. The phases of the overtones typically depend on the instrument and are partly correlated with the swinging of the keynote. This, together with the relation between the power of the overtones, produces the perceived sound of the instrument.

If the keynote has frequency f_0 , what are the frequencies of the overtones? This will depend on the type of instrument, but for string instruments, the overtones can be well represented¹ as

$$f_k = k f_0, \quad k = 1, 2, \dots$$

Task A [50]: Studying Trombone and Cello

Load the provided files **cello.mat** and **trombone.mat**. These files contain the time-domain signals of a tone played by a cellist and a trombonist at The Academic Orchestra in Sweden. You can listen to the tones by using the MATLAB command

$$\text{soundsc}(\text{cello}.x)$$

Import the data (**cello** or **trombone**) into MATLAB and estimate the frequency content using some appropriate method (note: the **.dt** parts of the data contain the sampling period T_s)

¹ It is worth noting that the stiffness of the string will actually produce some frequency offsets such that the overtones will not be exact multiples of the fundamental frequency. A more precise model of the overtones taking the string stiffness into account can be found as

$$f_k = k f_0 \sqrt{1 + B k^2}$$

where B is a positive stiffness parameter.

Examine the result using both a linear and logarithmic scale.

1. **How long are the recorded samples in time (seconds)?**
2. **What are the frequencies of the cello and trombone keynotes?**
3. **Do the overtones appear at integer multiples of the keynotes?**
4. **How many overtones can you see for the cello and the trombone sounds?** (Hint: use the logarithmic scale for magnitude).
5. These sounds were recorded using a really bad tape-recorder, and thus contain a lot of noise. **Can you see a strong noise peak at a particular frequency? What is the significance of that particular frequency?** (Hint: the tape recorder was not battery charged.)
6. Now assume that due to a buggy component in the recording system the cello recording got corrupted in the following way (note that the equation below represents the corrupting system)

$$\text{CelloCorrupted}[n] = n \times \text{cello}[2n + 1] + 5$$

- a. **Simulate the corrupted data.**
 - b. **Assuming that you have access only to the corrupted data and the corruption formula above, do you think it is possible to recover the original data?** (Justify your answer).
 - c. **If your answer to the above is Yes, then write a code to recover the original data.**
7. Similarly assume that the trombone recording got corrupted in the following way

$$\text{TromboneCorrupted}[n] = \frac{1}{2}(\text{trombone}[n] + \text{trombone}[n - 1])$$

- a. **Simulate the corrupted data. What has the buggy component done to the signal?**
- b. **Assuming that you have access only to the corrupted data and the corrupting formula above, do you think it is possible to recover the original data?** (Justify your answer).
- c. **If your answer to the above is Yes, then write a code to recover the original data.**

Task B[50]: Studying Three Instruments of Your Choice

Now make your own recording of any three musical instruments (one for each member of the group) and

- a. Analyze their frequency content and explore some interesting aspects in line with Task A (parts 1 to 7) ---- Individual level*
- b. Based on the five signal samples you have (02 from Task A and 03 from Task B), show what is similar, what is different, and what are the various interesting aspects of different music samples.*
- c. Discuss how a synthesizer could use the frequency information (keynote and overtones) to create the sound of the instrument artificially.*
- d. Discuss how a detection system could use this information to identify the presence of an instrument's sound in a recording.*